Introduction to the financial system

CASE STUDY: MORTGAGE TRADING ANALYSIS IN POWER BI

Nick Edwards
Capital Markets Analyst at Mynd

datacamp

Welcome!

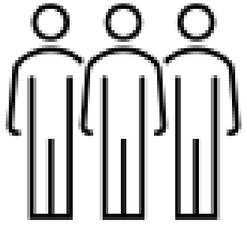
- Learn about the **financial system**
- You play the part of a **Junior Trader** at **Sooper Mortgage**
  - Fictional non-bank lender
  - United States based
- Execute a trade of mortgages in the **capital markets**
  - The date is September 21, 2021
  - Settlement date is October 13, 2021
- Multiple datasets that describe a population of over 5,000 mortgages!
  - Derived from real mortgage data
  - Derived from real market data

Prerequisites

- Test your skills and apply them to complex problems
  - Build confidence with new skills and strategies
- Comfortable with DAX, a bit of M, and the Power Query Editor
- Recommended courses:
  - Intermediate DAX in Power BI
  - Data Transformation in Power BI
- No banking or finance skills needed

Banks & Financial Institutions

Use deposits from savings accounts to buy investments which earn interest.



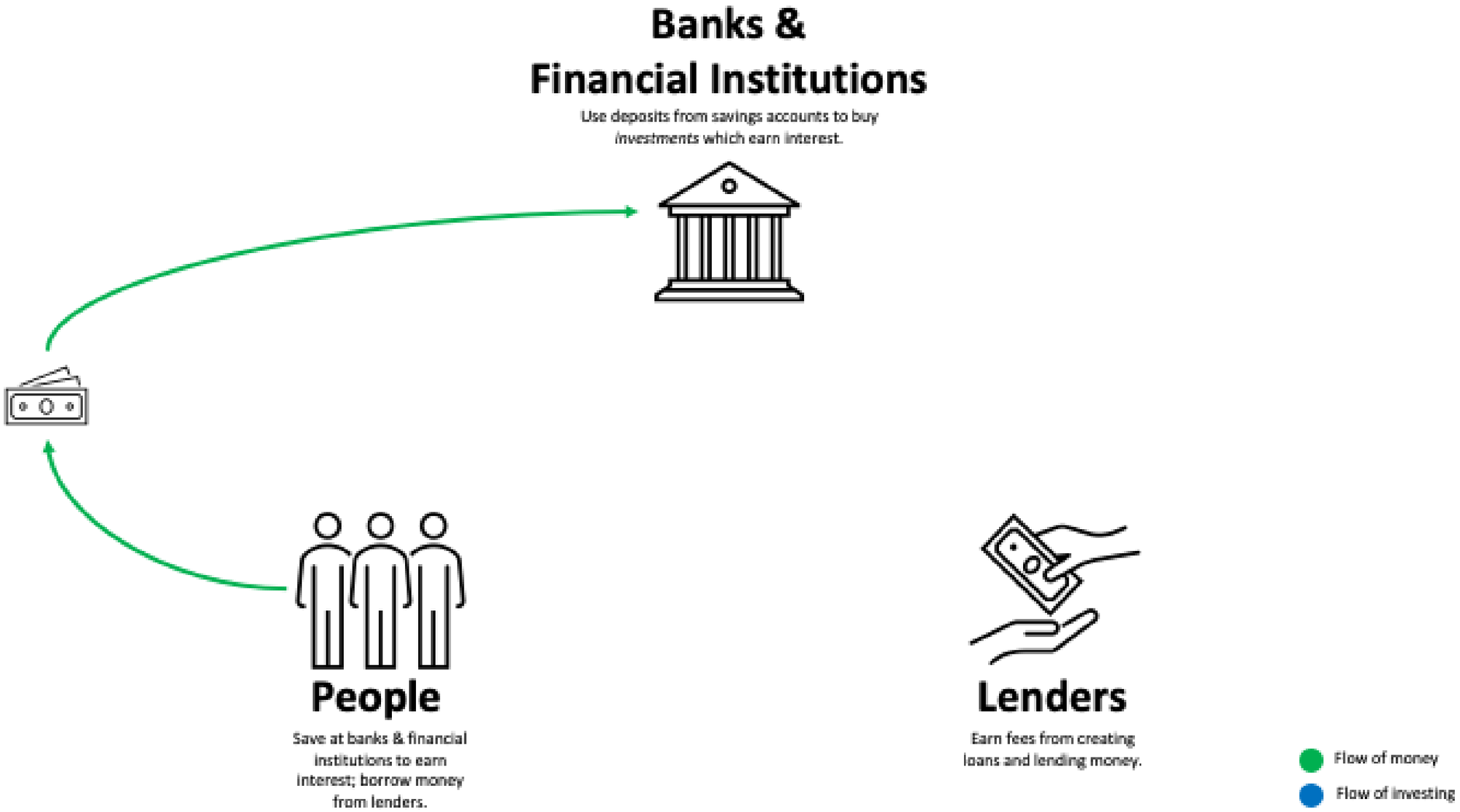
People

Save at banks & financial institutions to earn interest; borrow money from lenders.

Lenders

Earn fees from creating loans and lending money.

- Flow of money
- Flow of investing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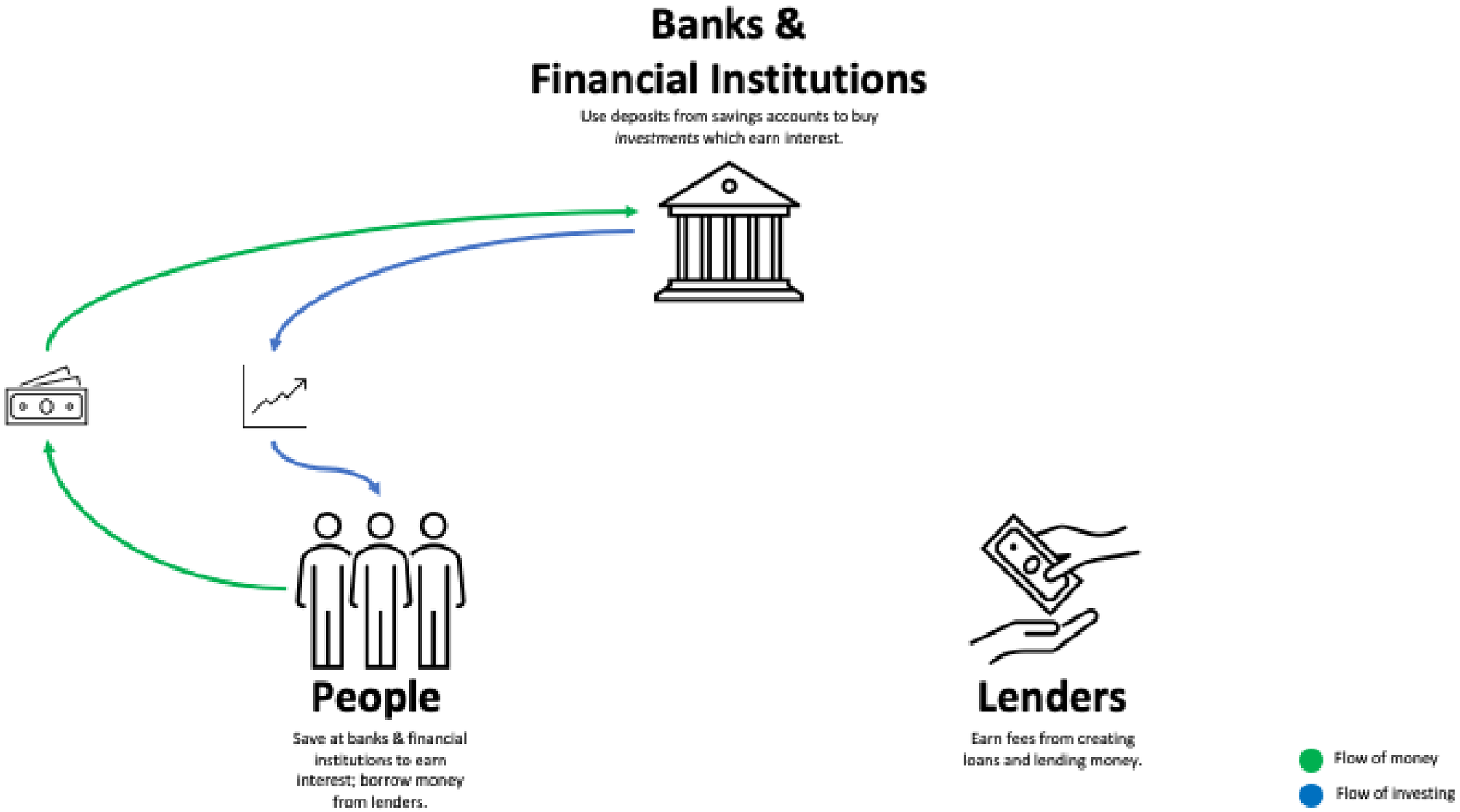
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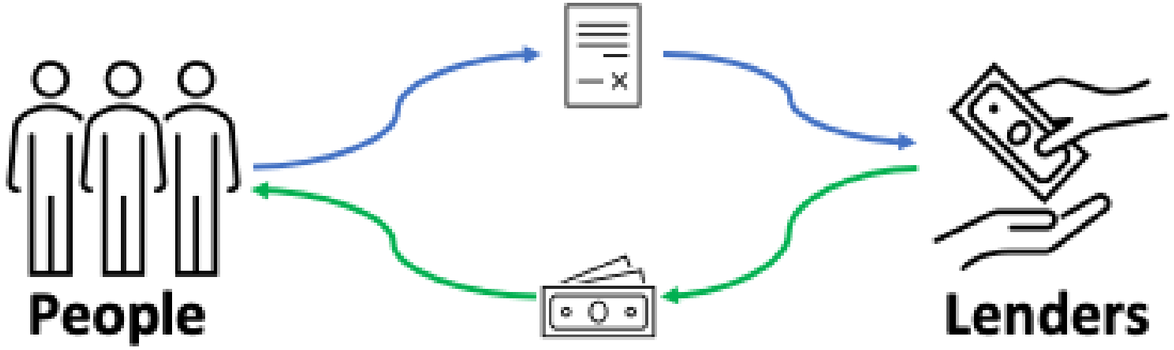

Flow of money

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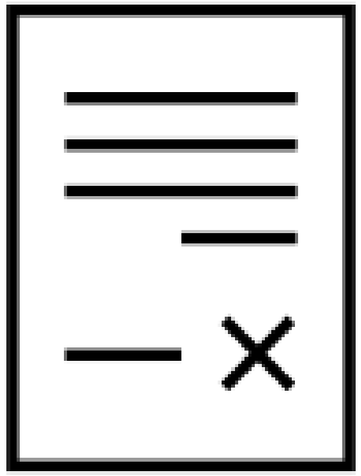

Loan Agreement

- Repay the money borrowed (usually with interest)
- Timely payments (usually monthly)

Banks & Financial Institutions

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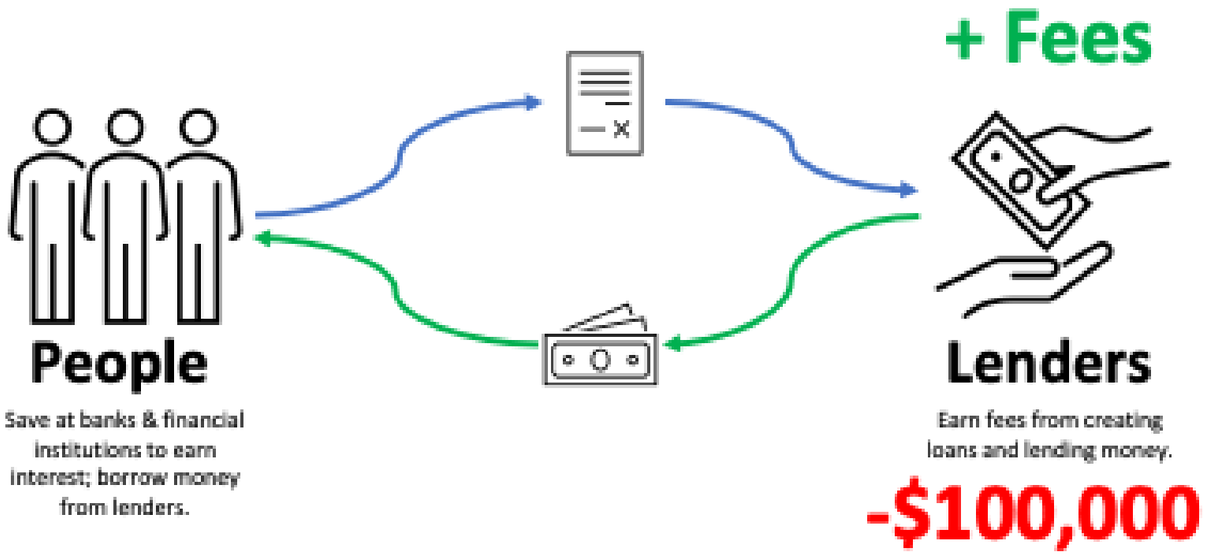

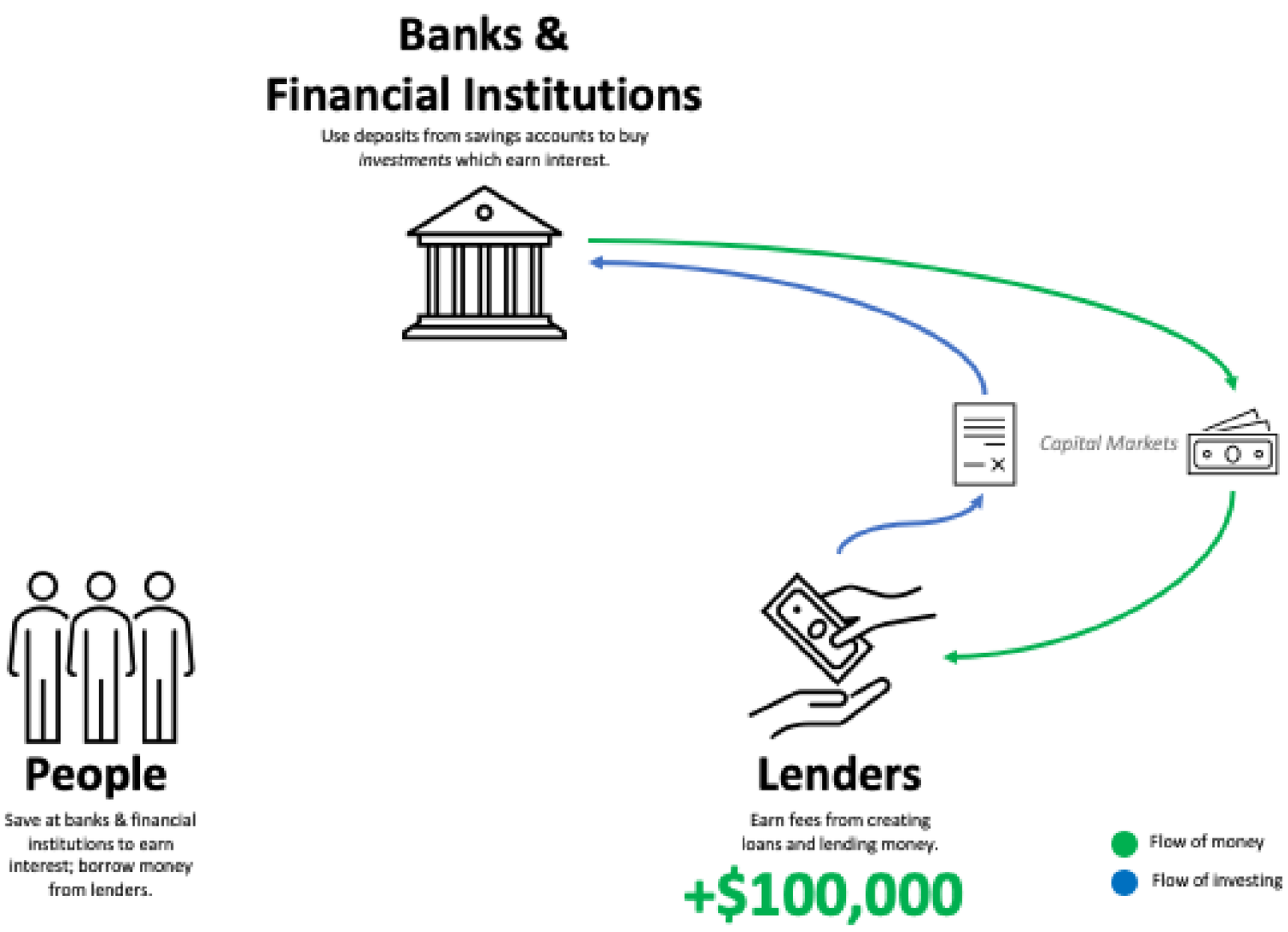
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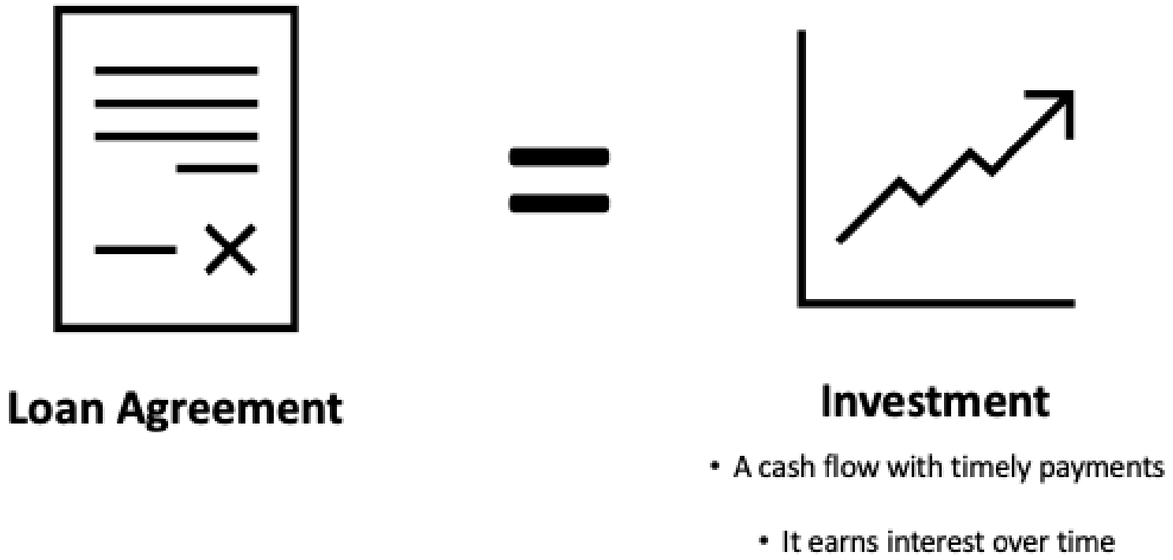
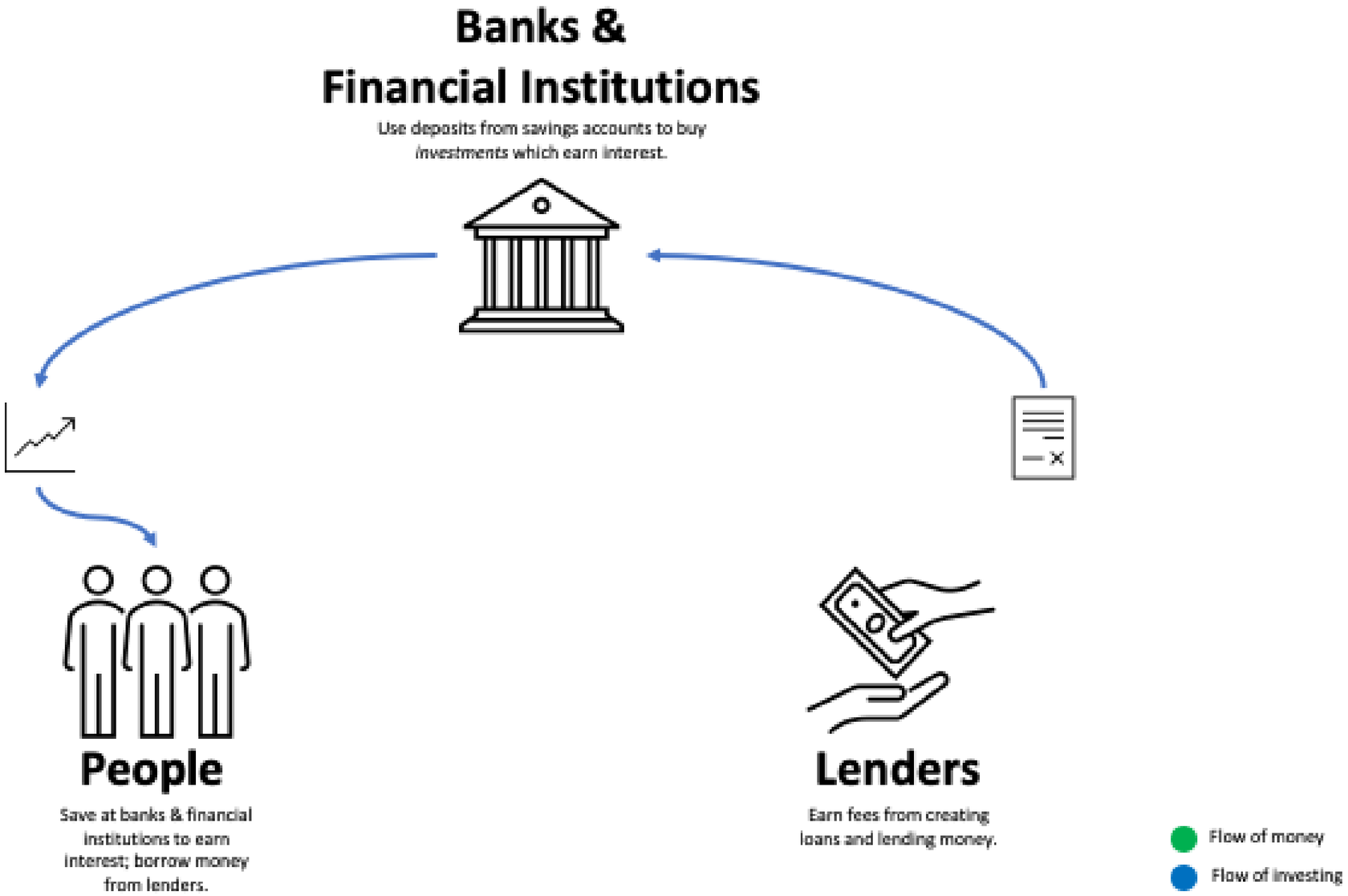
Use deposits from savings accounts to buy investments which earn interest.
Capital Markets
People
Save at banks & financial institutions to earn interest; borrow money from lenders.
Lenders
Earn fees from creating loans and lending money.
+$100,000
Flow of money
Flow of investing



- Example:
  - A **30YR** loan for `$100,000` at **5%** annually pays `$95,000` in interest over the full term.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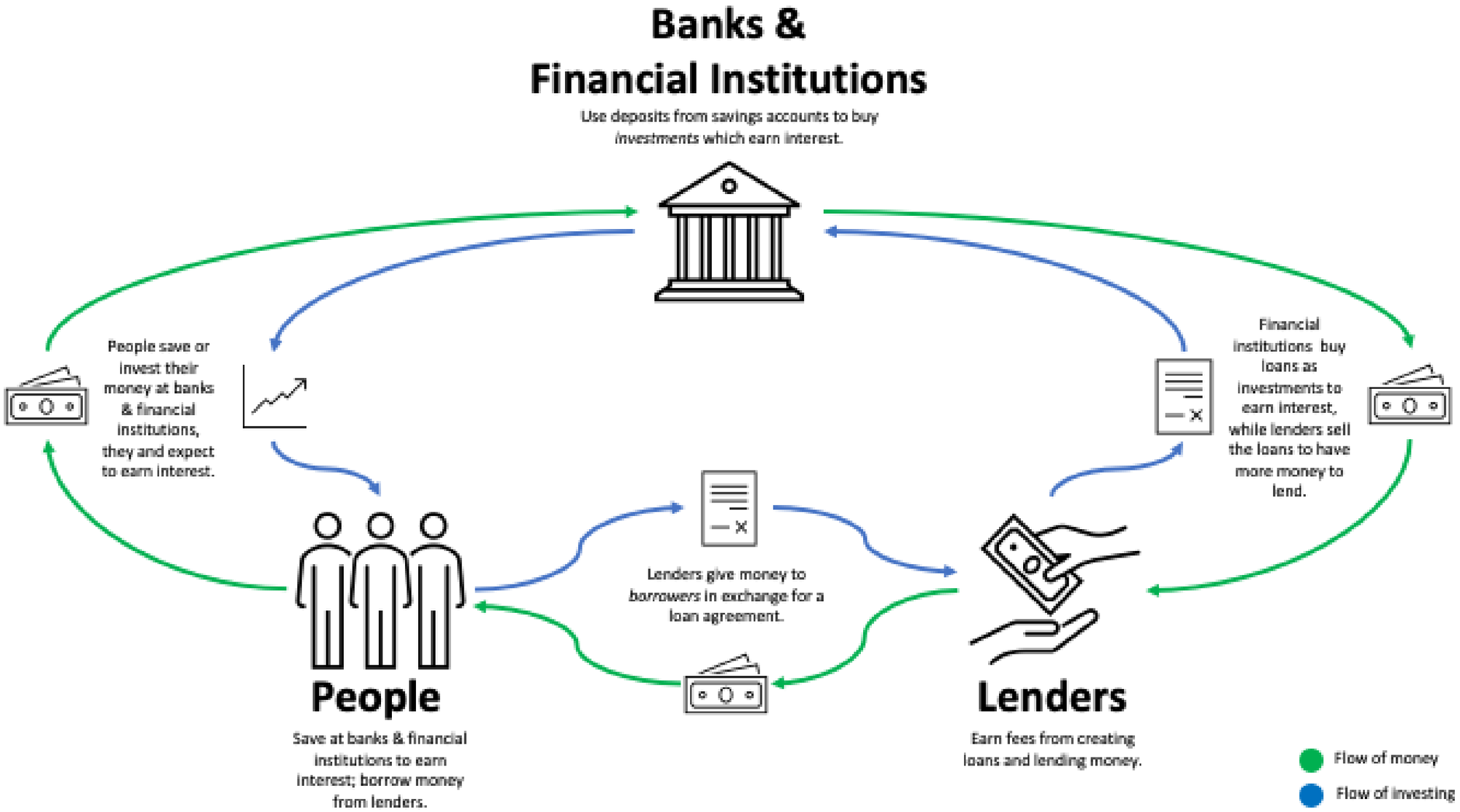
Banks &
Financial Institutions
Use deposits from savings accounts to buy investments which earn interest.
People save or invest their money at banks & financial institutions, they and expect to earn interest.
Financial institutions buy loans as investments to earn interest, while lenders sell the loans to have more money to lend.
Lenders give money to borrowers in exchange for a loan agreement.
People
Save at banks & financial institutions to earn interest; borrow money from lenders.
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Earn fees from creating loans and lending money.
Flow of money
Flow of investing

Let's practice!

CASE STUDY: MORTGAGE TRADING ANALYSIS IN POWER BI

What is a mortgage?

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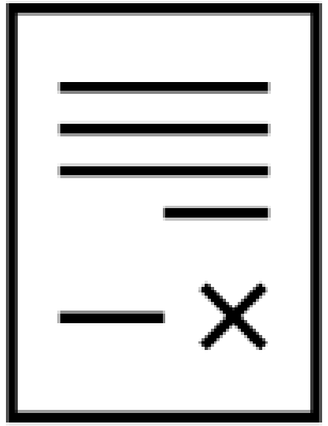
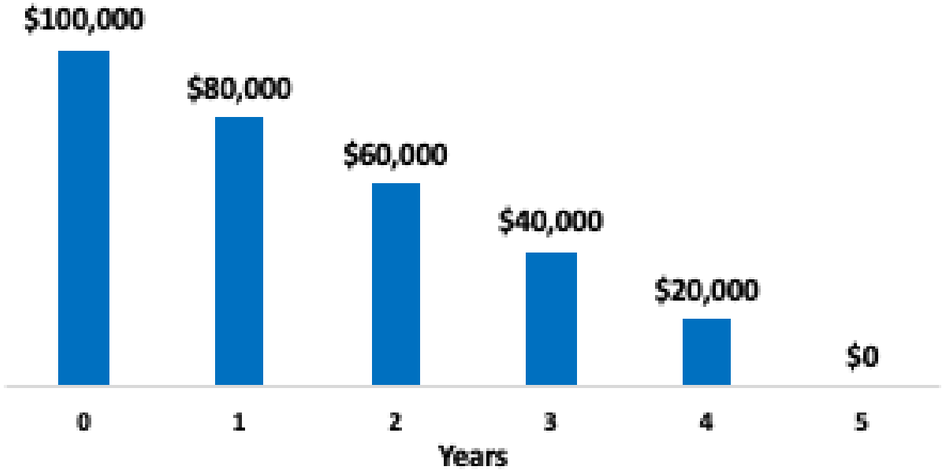
What is a loan agreement?

A **loan agreement** is the borrower's promise to repay the money to the lender.

- Essentially a repayment plan



Loan Agreement

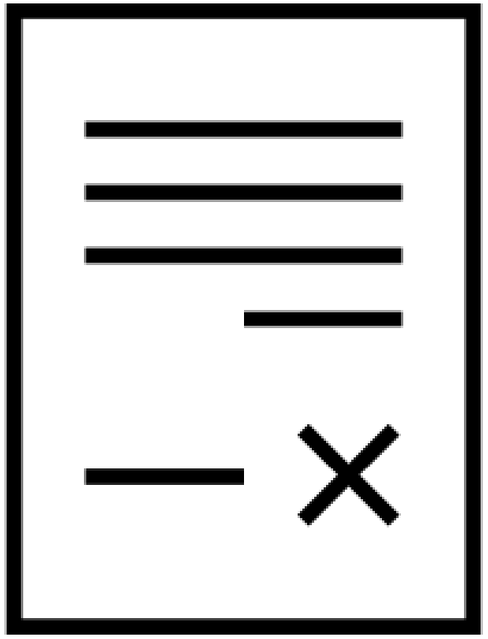


Repayment plan

What is a loan agreement?

Parts of a loan agreement:

- Loan amount
- Payment
  - Principal
  - Interest

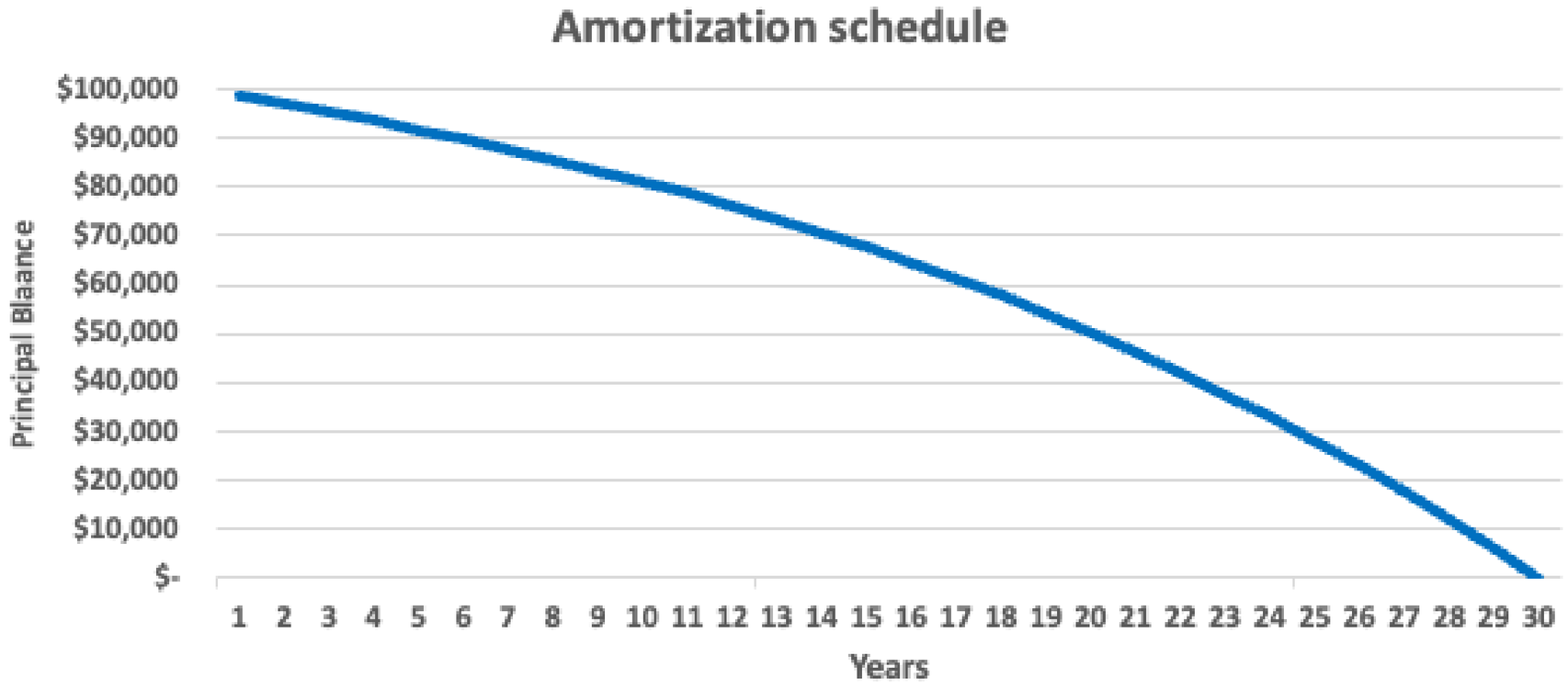


Loan Agreement

[1] https://www.investopedia.com/terms/m/mortgage.asp

Principal

- Principal is the current balance on the loan



Interest

- The **interest rate** is the proportion of interest owed on the principal balance each period.
  - Quoted as an annual rate

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Example

What is the interest due this month on a principal balance of $100,000 at a 5% annual rate?

Interest rate	Monthly interest rate	Principal Balance	Interest owed
5%		$100,000	

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5%	5% ÷ 12 = 0.417%	$100,000	

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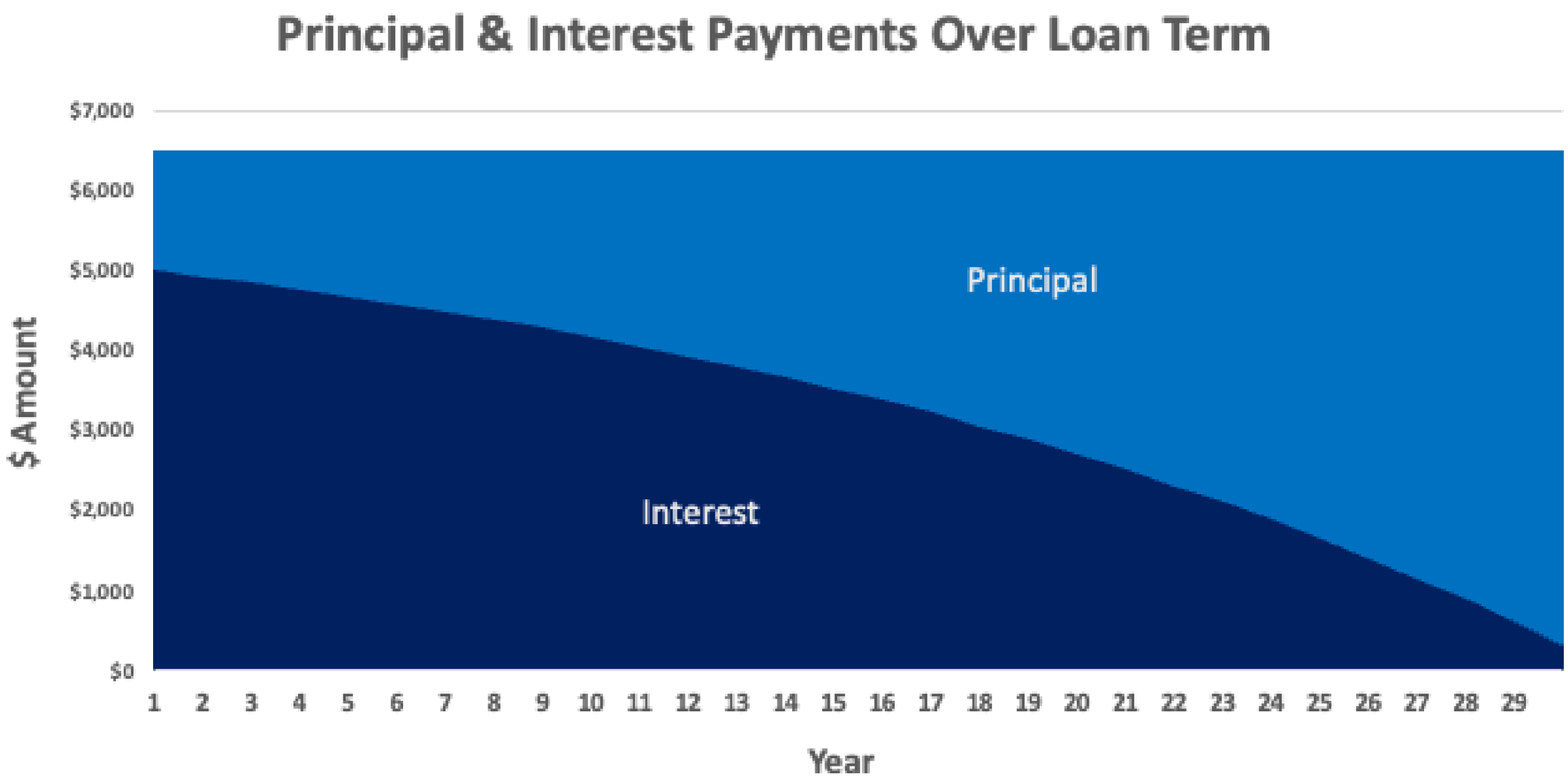
Example

What is the interest due this month on a principal balance of $100,000 at a 5% annual rate?

Interest rate	Monthly interest rate	Principal Balance	Interest owed
5%	5% ÷ 12 = 0.417%	$100,000	0.417% x 100,000 = $417

Amortizing


Principal & Interest Payments Over Loan Term
$7,000
$6,000
$5,000
$4,000
$3,000
$2,000
$1,000
$0
$ Amount
Principal
Interest
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29
Year

Amortizing in Power BI

Syntax

```
PPMT(<rate>, <per>, <nper>, <pv>[,<fv>,<type>])
```

- **This produces a negative value!**
- **The rate needs to be adjusted!**

Example: Find the principal amount on the **first payment** of a loan with a balance of **$100,000** with a **5% annual** interest rate and a **360** month term.

`PPMT(.05/12, 1, 360, 100000)` = `($-120.15)`

Click here to review the syntax in further detail

[1] https://learn.microsoft.com/en-us/dax/ppmt-function-dax

Amortizing in Power BI

2 Steps to Amortize:

1. Use `PPMT()` to find the **principal** portion of a payment
2. Subtract the **principal** amount from the current loan balance

Example continue...

```
[Principal Balance] + [PPMT]
```

```
[$100,000] + [-$120.15]
```

```
$99,879.85
```

Mortgage qualification

- **Credit score**
  - Missed payments reduce score as they show risk
  - Ranges from 300 (lowest) to 850 (perfect)
- **Debt-to-income (DTI) ratio**
  - `[Monthly debt payments] ÷ [Monthly income]`
  - Generally needs to be under 50%
- **Loan-to-value (LTV) ratio**
  - `[Loan amount] ÷ [Propert value]`
  - The lender needs to know they can get their money back in case of foreclosure
  - The lower, the better

[1] This is for educational purposes only. Consult your lender for actual terms and qualification guidelines.

Let's continue!

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